



SVP-25-031

10 CFR 50.73

June 4, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-29
NRC Docket No. 50-254

Subject: Licensee Event Report 254/2025-004-00 "Loss of Suppression Chamber Vacuum Breaker Safety Function due to Configuration Control Error"

Enclosed is Licensee Event Report 254/2025-004-00 "Loss of Suppression Chamber Vacuum Breaker Safety Function due to Configuration Control Error".

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of systems needed to mitigate the consequences of an accident, and 10 CFR 50.73(a)(2)(i)(B), any operation or condition prohibited by Technical Specifications.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Conner Bealer at 779-231-6207.

Respectfully,

A handwritten signature in black ink, appearing to read "Doug Hild", written over a faint circular stamp.

Doug Hild
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Quad Cities Nuclear Power Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00254

3. Page

1 OF 4

4. Title

Loss of Suppression Chamber Vacuum Breaker Safety Function due to Configuration Control Error

5. Event Date

Month	Day	Year
04	05	2025

6. LER Number

Year	Sequential Number	Revision No.
2025	004	00

7. Report Date

Month	Day	Year
06	04	2025

8. Other Facilities Involved

Facility Name
N/A☐ 050Docket Number
N/AFacility Name
N/A☐ 052Docket Number
N/A

9. Operating Mode

1

10. Power Level

100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Richard Swart – Regulatory Assurance

Phone Number (Include area code)

309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
N/A					N/A				

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 04/05/2025 at 04:00, Operations determined that a mispositioned manual isolation valve made both trains of Reactor Building to Suppression Chamber Vacuum Breakers simultaneously inoperable. The vacuum breaker inoperability could have prevented fulfillment of primary containment function following a design-basis accident with no operator intervention. The valve was not properly repositioned following refuel outage activities in March of 2025.

The cause of the valve mispositioning was inadequate standards in Procedure Use and Adherence during refuel outage activities. Immediate corrective action was to open the isolation valve. Subsequent actions will be to make the valve a locked-open valve, to improve related procedures, and to provide training related to this event.

This is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of systems needed to mitigate the consequences of an accident, and 10 CFR 50.73(a)(2)(i)(B), any operation or condition prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1	☒ 050	2. DOCKET NUMBER 00254	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 004	REV NO. - 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric – Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Loss of Suppression Chamber Vacuum Breaker Safety Function due to Configuration Control Error

A. CONDITION PRIOR TO EVENT

Unit: 1	Event Date: May 5, 2025	Event Time: 0400 hours CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100%

No systems, structures, or components that were inoperable at the start of the event contributed to the event.

B. DESCRIPTION OF EVENT

On 04/05/2025 at 04:00, Operations discovered that manual isolation valve [V] 1-1601-84B, U1 Torus Line 1-1635B-1/2 inch LA Root Valve, was in the closed position. Closure of the 1-1601-84B (-84B) valve prevents operation of air operated Reactor Building to Suppression Chamber Vacuum Breakers [VACB][BF] 1-1601-20A and 1-1601-20B, hereafter denoted as "the vacuum breakers." The -84B valve isolated differential pressure sensors [DPS] from the suppression pool atmosphere, subsequently preventing the operation of solenoid valves [PDSV] providing air to the vacuum breakers. Technical Specification (TS) 3.6.1.7, Reactor Building to Suppression Chamber Vacuum Breakers, Conditions C and E were entered for each vacuum breaker. Condition C is for one line with one or more reactor building to suppression chamber vacuum breakers inoperable for opening; Condition E is for two lines with one or more vacuum breakers inoperable for opening. The -84B valve was opened, and TS 3.6.1.7 was exited on 04/05/2025 at 0400. The vacuum breaker inoperability could have prevented fulfillment of primary containment function following a design-basis accident with no operator intervention. Additionally, the TS 3.6.1.7 Condition E Completion Time of 1 hour was not met based on the Required Action to restore vacuum breakers in one of two lines to operable status following entry into Mode 2. Also, TS Limiting Condition for Operation (LCO) 3.0.4 was not met when Mode 2 was entered without any specific allowances or risk assessments performed for the vacuum breaker inoperability related to TS 3.6.1.7.

During the refuel outage, March 10, 2025, through April 1, 2025, Local Leak Rate Tests (LLRTs) and valve maintenance activities included closure of the -84B valve. At the conclusion of the as-left LLRT on 03/28/25, steps to open the -84B and other valves were annotated that they were to be left closed per the Unit Supervisor, with a Senior Reactor Operator (SRO) having an expectation that subsequent outage clearance order activities would later recover the desired valve position. No peer reviews were performed on the incomplete steps, nor were any verifications made related to the applicable clearance order, resulting in the isolation valves remaining closed the duration of the refuel outage.

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1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1	☒ 050	2. DOCKET NUMBER 00254	3. LER NUMBER				
	☐ 052		<table border="1"><thead><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr></thead><tbody><tr><td>2025</td><td>- 004</td><td>- 00</td></tr></tbody></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025
YEAR	SEQUENTIAL NUMBER	REV NO.					
2025	- 004	- 00					

NARRATIVE

The valve remained closed through reaching the Mode of Applicability for TS 3.6.1.7, Mode 2, on 03/31/2025 at 1052, and through Mode 1 operation. In subsequent days, unexpected continuous operation of a compressor [CMP] associated with differential pressure control between the drywell and the suppression chamber resulted in troubleshooting of compressor control instruments. This is when the -84B valve was found closed. The -84B valve was opened on 04/05/2025 at 0400. The total time the vacuum breakers were inoperable in the modes of applicability was 4 days, 17 hours, and 8 minutes. TS 3.6.1.7 Condition C has a 7-day completion time, which was not exceeded. TS 3.6.1.7 Condition E has a 1-hour completion time, which was exceeded.

Event Notification System report 57647 was submitted on 04/05/2025 at 1133.

C. CAUSE OF EVENT

The cause of the self-identified event was inadequate standards in Procedure Use and Adherence.

The procedure step to recover the -84B has instructions to "perform system restoration as follows, or as directed by the Unit Supervisor." The SRO performing the LLRT procedure incorrectly assumed that the -84B would be restored through some other process but did not validate this assumption or obtain a peer check, as was required by procedure. Instead, the procedure step was annotated that the valve would be left closed per the Unit Supervisor.

A failure to properly reinforce the required Human Performance and Technical Human Performance standards contributed to this event.

D. SAFETY ANALYSISSystem Design

Automatic vacuum relief devices on the drywell and the suppression chamber prevent the primary containment from exceeding the design external-to-internal pressure differential. The drywell is designed for a maximum external pressure of 2 psi greater than the concurrent internal pressure. The suppression chamber is designed for a maximum external pressure of 2 psi greater than the concurrent internal pressure based on the original design calculations; however, the overpressure capability of the suppression chamber is conservatively stated to be 1.0 psi.

The suppression chamber vacuum breakers prevent excessive vacuum in the suppression chamber relative to the reactor building by admitting reactor building air at a preset pressure differential that does not exceed the equivalent of 0.5 psid. Two vacuum breaker valves in series are used in each of two lines leading from the reactor building atmosphere. One valve in each line (vacuum breakers 1-1601-20A or -20B) is air-operated and actuated by a differential pressure signal, independently of electrical power. The second valve in each

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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 004	REV NO. - 00

NARRATIVE

line is self-actuating. The combined pressure drop at rated flow through both valves does not exceed the difference between suppression chamber design external pressure and maximum atmospheric pressure.

Safety Impact

There were no safety consequences as a result of this event. There was no radiation release associated with this event. Operations took immediate actions upon discovery of the condition. The vacuum breakers will fail open on the loss of electrical control power. Operator actions could also address this condition in an applicable design-basis scenario.

This event is considered a safety system functional failure per NEI 99-02.

E. CORRECTIVE ACTIONS**Immediate:**

1. Valve 1-1601-84B was returned to the open position.

Follow-up:

1. New default position for the valve will be 'Locked Open.'
2. Startup procedures will be revised to verify the position of the valve.
3. Training will be provided for the Operations Department related to this event.

F. PREVIOUS OCCURENCES

The station events database, LERs and IRIS were reviewed for similar events at Quad Cities Nuclear Power Station. Two station LERs in the last three years listed below include events where written instructions were not properly recognized or followed by Operations, resulting in TS violations.

LER (254/2024-01-00) Technical Specification 3.5.2 Action Not Performed Due to Inadequate Procedure Adherence, 1/21/25. Procedure adherence related issue resulted in a Reactor Pressure Vessel Water Inventory Control (RPV WIC) drainage event. Associated with that event was a failure to recognize the need to enter RPV WIC related TS Actions.

LER (254/2025-001-00) Technical Specification 3.6.4.2 Not Performed Due to Tagout Boundary Placement Error, 3/7/25. Failure to implement the administrative controls in a tagout resulted in a secondary containment isolation valve being left in an open position.

G. COMPONENT FAILURE DATA

There were no equipment failures associated with this event.

This event has been reported to IRIS.